CardioGuard AI

Early CVD & Atrial Fibrillation Detection Systems

Generated on: 2026-07-22 | Author: AI Pair Programming Agent

1. Project Overview & Present Status

CardioGuard AI is a high-fidelity web and mobile application designed to detect early-stage Cardiovascular Disease (CVD) and Atrial Fibrillation (AF) risk. The application compiles correctly, and the developer build is serving locally.

- **Vite Dev Server Status:** ACTIVE (Running on http://localhost:5173/)
- **Production Web Build:** SUCCESSFUL (tsc compiler checked, zero errors)
- **Capacitor Integration:** COMPLETED (Gradle wrapper configured in /android)
- **Biological Signal Peak Detection:** OPERATIONAL (Parses raw ECG CSV inputs)
- **HRV Metric Calculators:** OPERATIONAL (Real-time SDRR & RMSSD equations)
- **Simulated CNN-LSTM Classifier:** OPERATIONAL (Rule-based logits classifier)

2. Core Completed Features

Raw ECG QRS Peak Detector
Upgraded CSV parser detecting localized R-peaks in raw amplitudes with an adaptive 65% signal threshold and a 250ms refractory period block.

Biometric Analytics (SDRR & RMSSD)
Real HRV calculations executing standard deviation (SDRR) and root mean square successive differences (RMSSD) on beat intervals to diagnose cardiac patterns.

Vitals Sync Simulator (Google Fit)
Connected telemetry synchronization dashboard syncing 12 hourly heart rate logs to the line chart overlaying normal HRV baselines.

ECG Lead-II Waveform Oscilloscoope
Renders dynamic SVG waves displaying biological cardiac characteristics (regular P-waves and QRS structures vs rapid baseline f-waves and chaotic RR intervals).

CNN-LSTM Deep Learning Visualizer
Interactive visual layer block schematic illustrating Conv1D feature filters (spatial shape extraction) and Bidirectional LSTM gates (temporal sequence modeling).

Mobile-Responsive UX Layout
Structured media breakpoints ensuring panels collapse cleanly on smaller phone screens, featuring persistent results export logs.

3. Project Directory Tree Structure

```
D:\cardiogaurd
% % %  public/                    # Client static assets (favicons, manifest)
% % %  src/
%    % % %  components/
%    %    % % %  HeartRateChart.tsx   # Custom SVG chart & tooltip coordinates tracking
%    %    % % %  GoogleFitConnector.tsx # Mock auth modal and data telemetry syncing
%    %    % % %  DatasetUploader.tsx  # Raw ECG peak detector & preset wave parser
%    %    % % %  DetectionRunner.tsx  # AI analysis sequencer & ECG wave generator
%    %    % % %  ResultsView.tsx      # Diagnostic radial gauge & ECG Lead-II waveform simulator
%    %    % % %  CnnLstmArchitecture.tsx # TF Lite CNN-LSTM network layer schematic
%    % % %  App.tsx                  # Core React context layout & state coordinators
%    % % %  App.css                  # Local stylesheet overrides
%    % % %  index.css                # Global glassmorphism, responsive grids, and typography
%    % % %  main.tsx                 # StrictMode app renderer entry point
% % %  android/                    # Native Android workspace wrapping output files
% % %  capacitor.config.ts         # Android wrapper packaging configuration specifications
% % %  package.json                # Project script targets and Capacitor CLI packages
% % %  README.md                   # Setup requirements & manual validation specifications
% % %  CardioGuard_Project_Report.pdf # Project development summary report (Current File)
```

4. Native Android APK Build Guide

To package and compile this project workspace into a native Android .apk package for device installation, execute the following commands in the workspace root:

1. Build production web bundle

```
npm run build
```

Compiles source TSX files into the static /dist directory.

2. Synchronize web assets

```
npm run cap:sync
```

Copies web assets from /dist into native Android public folders.

3. Open in Android Studio

```
npm run cap:open
```

Opens the wrapper Gradle workspace directly inside Android Studio.

Once Android Studio Gradle indexing completes, navigate to:
Build -> Build Bundle(s) / APK(s) -> Build APK(s) to compile app-debug.apk.